Supplementary Materials for: "Higher-order shortest paths in hypergraphs"

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S1. EFFECT OF TIME WINDOW ON AGGREGATION

In this section, we describe our approach to time-window aggregation for temporal hypergraphs. For static systems, we first show how neglecting temporal information results in a gradual loss structural information. We do this by constructing static graphs from successively larger portions of the entire Copenhagen dataset. In Figure S1 we plot path-length distributions for networks constructed using the first $k\%$ of total timestamps available, with k ranging between 0.1% and 20%. For each case, we also randomise the data by making use of the higher-order configuration model. We observe that the distribution shift becomes less and less pronounced and is almost indistinguishable by the time we reach an aggregation over 20% of the total available dataset.

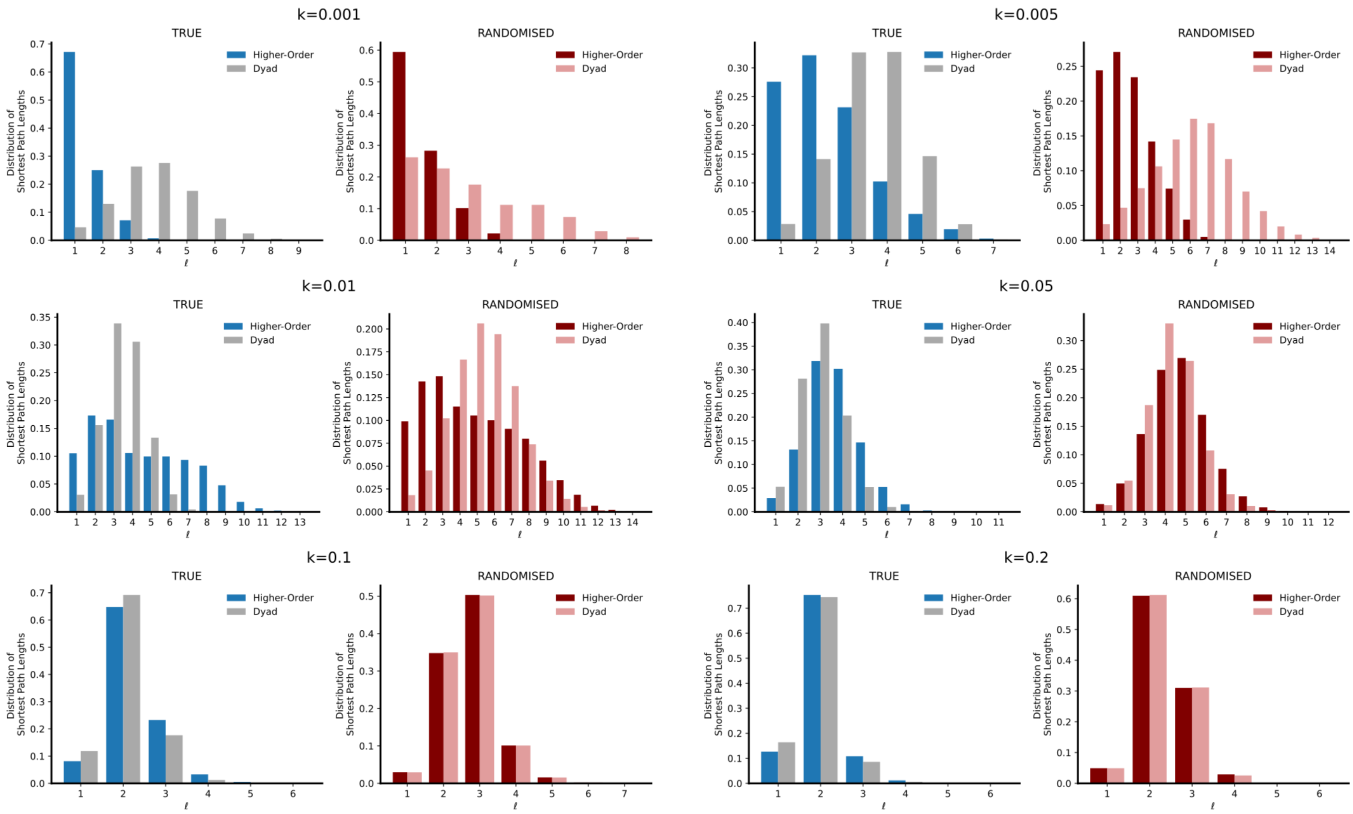


FIG. S1: Influence of aggregation on datasets

For temporal systems it is often necessary to aggregate intermediate time-windows to avoid the situation in which successive snapshots are so sparse that one cannot extract meaningful connectivity patterns. As mentioned in the Methods, for temporal Friends & Family data we use a data-driven approach to choose the appropriate size of the aggregation window. We select the smallest time window for which the size of the largest connected component (LCC) stabilises. Figure S2 provides 3 statistics related to the LCC as a function of the aggregation window, over 3 different days. We select 8 hours (480 minutes) as a suitable aggregation window, yielding 3 snapshots per day and thus between 84-99 snapshots per calendar month of data. To ease the computational load, we additionally divide the Friends & Family dataset into subsets that each span a single calendar month. This yields 10 separate datasets which span the period of August 2010 and May 2011.

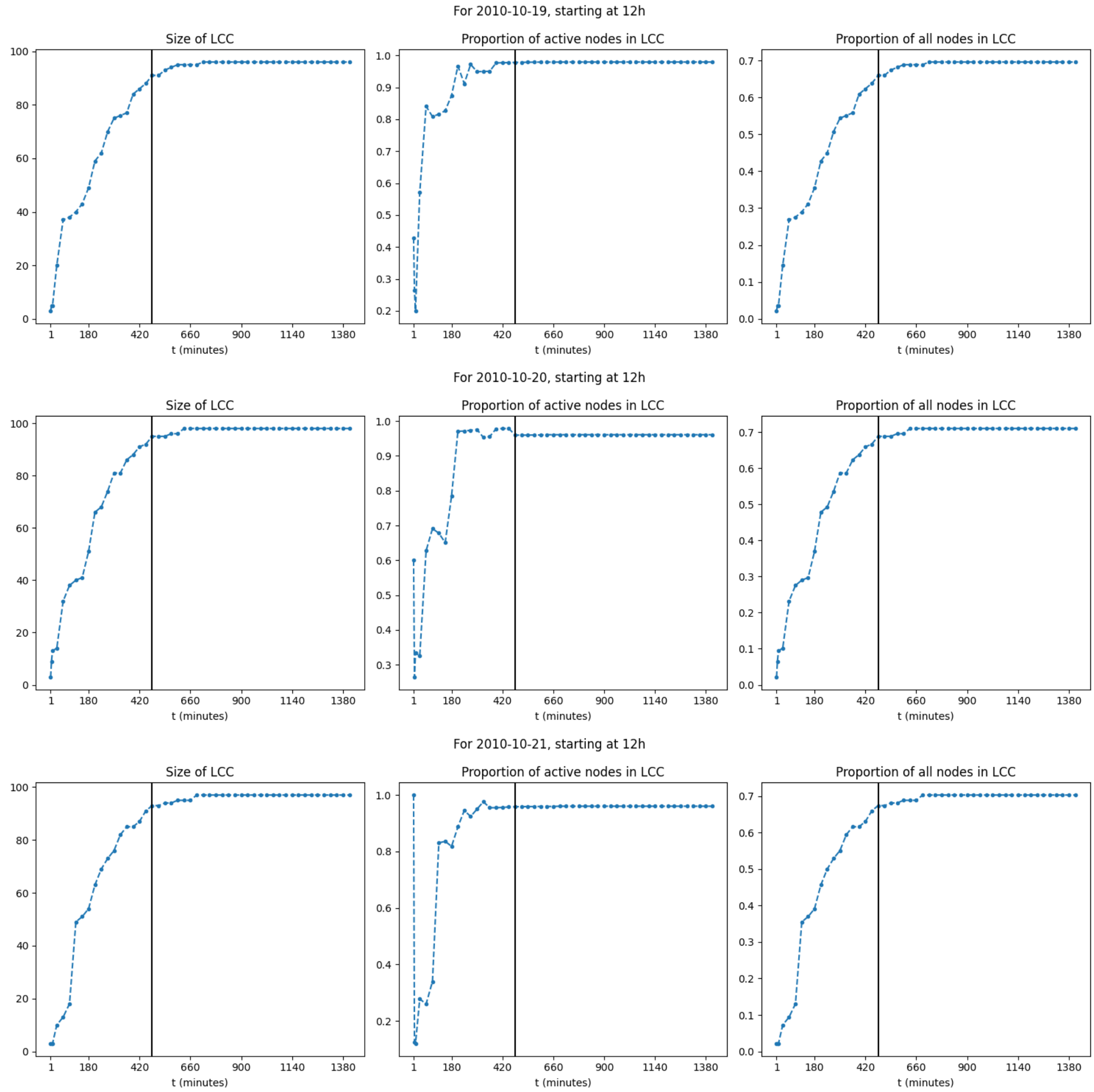


FIG. S2: Selecting an aggregation window for the Friends & Family dataset. Rows represents different days. The first column plots the size of the LCC (in number of nodes) as a function of the size of the aggregation window in minutes (t). The second column gives the proportion of active nodes that also participate in the LCC. The final column plots the proportion of all nodes (both active and inactive) forming part of the LCC.

S2. PATH LENGTH DISTRIBUTIONS

In this section, we plot path length distributions for all datasets considered, thus providing more complete versions of Figures 2 and 5 from the main text.

We first supplement Figure 2 of the main text by plotting path length distributions in the static case for all datasets in Figure S3. We provide distributions of path lengths for the entire dataset (in dark blue) and pure pairwise (in grey). Furthermore, we now also plot the distributions for networks constructed by considering only those interactions which are truly higher-order (in light blue). We observe that all systems display a very low diameter. Additionally, we observe that the path length distributions are very similar for many of the datasets, displaying only slight shifts. This contrasts drastically when compared to the later temporal, case.

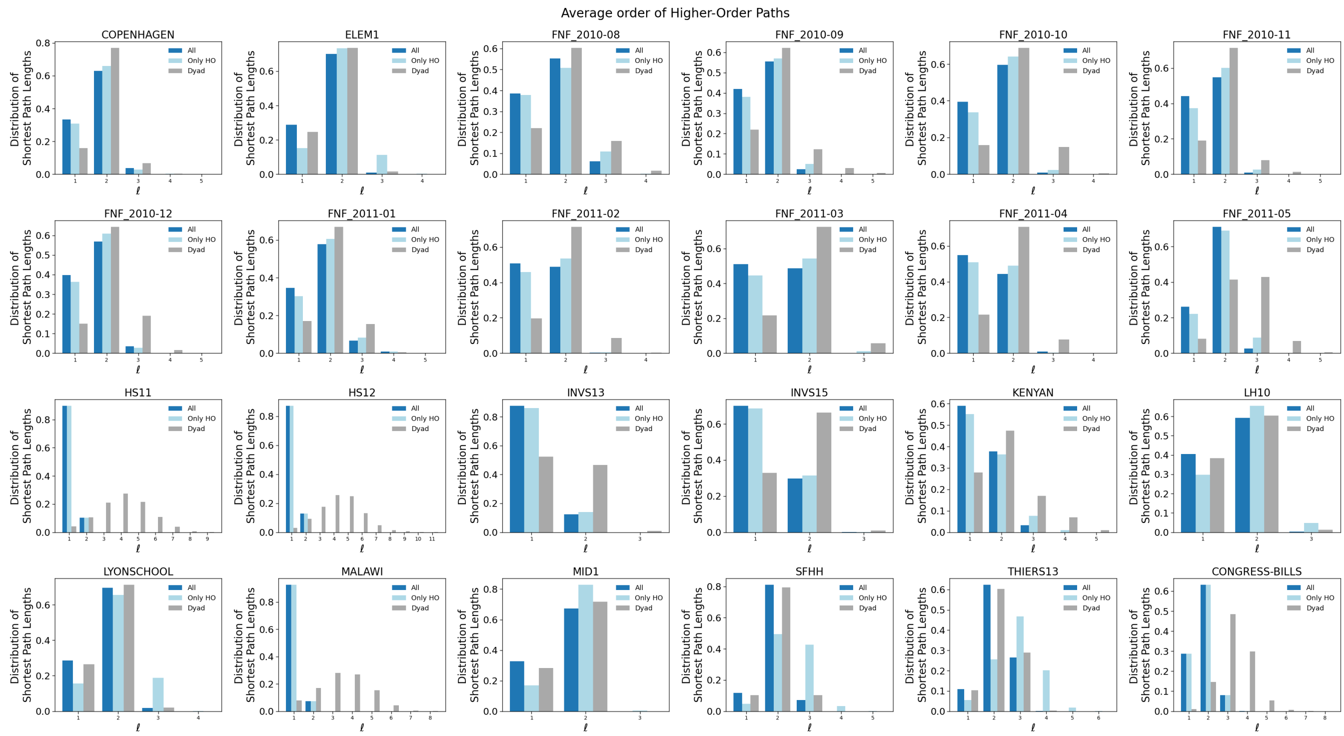


FIG. S3: Shortest paths for the static hypergraph of all datasets. Path-length distributions for the entire dataset (dark blue), for only those truly higher-order interactions (light blue), and for purely pairwise interactions (grey) for static hypergraphs.

Next, we plot in Figure S4 the proportion of paths in static hypergraphs which exist only as dyadic paths. (This is the static analogue to Figure 6a of the main manuscript). We see here that for most datasets none of the paths disappear, which is to be expected given the temporal aggregation that has taken place.

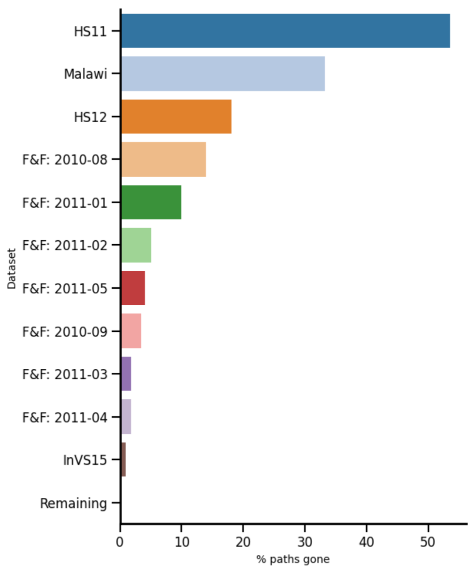


FIG. S4: Percentage of paths which consist solely of pure pairwise interactions and thus disappear when considering only the system composed of higher-order interactions.

Next we complete Figure 5 of the main text by providing path fastest duration distributions for all datasets in Figure S5. As in the static case, we now also plot the the path length distributions of the networks constructed by considering only those interactions which are truly higher-order (in green). Here we observe the distribution shifts discussed in the main text much more clearly, and the influence of the pure pairwise interactions on the system diameter is much more marked. However, the observations are also dataset-dependent. For some datasets, such as Thiers13 which describes interactions at school, we have almost no higher-order interactions, whereas for others, such as HS11 & HS12 - also interactions at high schools - no pure pairwise interactions are present.

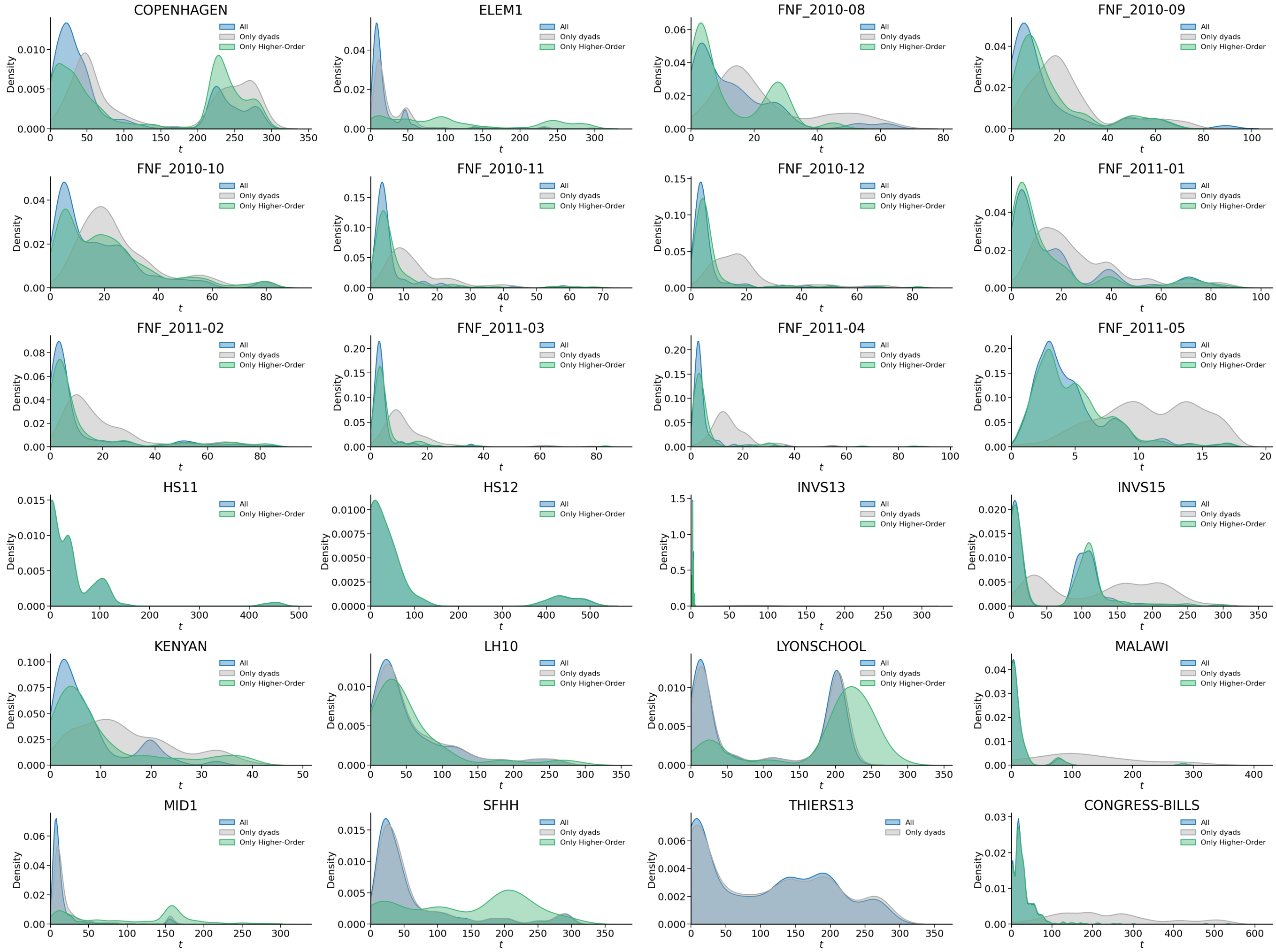


FIG. S5: Distributions of fastest path durations for all temporal datasets in blue (all interactions), green (only higher-order) and grey (only pairwise).

S3. ALTERNATIVE DEFINITIONS FOR AVERAGE PATH SIZE

In this section, we study the impact of defining average hyperpath size in alternative ways. Recall that average path size is defined in the main text as the average size of all edges traversed in a particular hyperpath. Whenever multiple hyperedges are present at a certain step, we resolve ambiguity by selecting the hyperedge of minimum size. It is, however, also possible to use a different selection strategy, and here we study average hyperpath size when selecting either the mean or the maximum of the possible hyperedges.

Figure S6 extends Figure 2c of the main text by plotting the average path size across all datasets for 3 different strategies. We observe that the average path size for the case of minimum, mean and maximum exhibits similar behaviour across all datasets. In addition, we observe the same trends of decreasing size for increasing path length across almost all of the datasets.

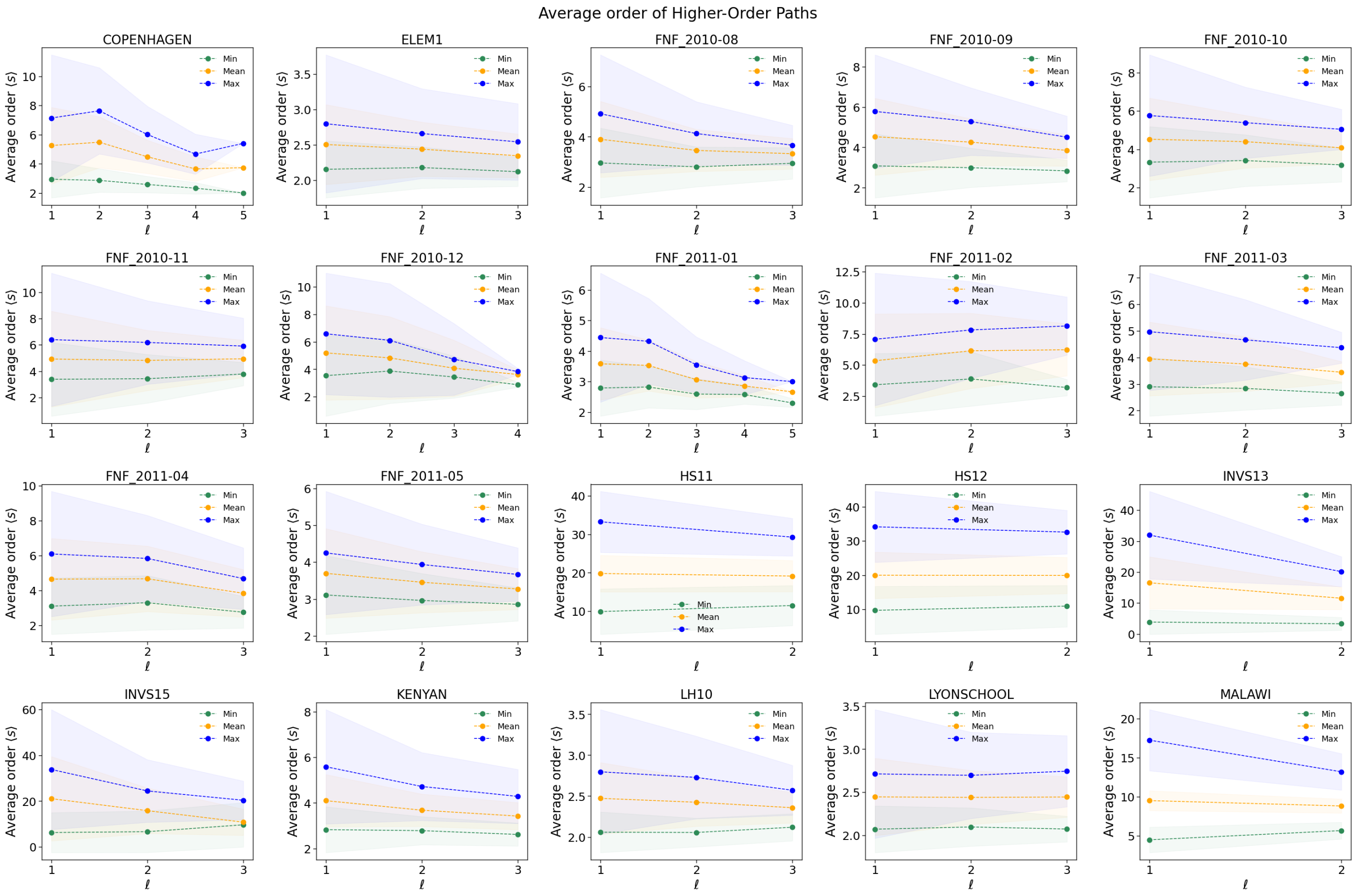


FIG. S6: Average path size $\langle s \rangle$ of the higher-order shortest paths as a function of the path length across all datasets when using 3 different strategies (minimum in green ($\langle s \rangle_{\text{min}}$), mean in orange $\langle s \rangle_{\text{mean}}$, maximum in blue $\langle s \rangle_{\text{max}}$) to resolve ambiguities.

To further study the impact of alternative strategies on average path size, we calculate how much the metric changes when using different hyperedge selection strategies to resolve ambiguities. For the static case, in Table S1 we first calculate average path size using the minimum-size hyperedge. We then determine how much average path size increases when choosing respectively the mean and the maximum hyperedge by calculating the proportion $\langle s \rangle_{\text{max}}/\langle s \rangle_{\text{min}}$ and $\langle s \rangle_{\text{mean}}/\langle s \rangle_{\text{min}}$. We observe here that regardless the choice of strategy, most datasets exhibit almost no increase in size. However the increase in path size can be up to 8 times larger, such as for the InVS15 workplace dataset.

Datasets	$\langle \mathbf{s} \rangle_{\min}$	$\langle \mathbf{s} \rangle_{\text{mean}} / \langle \mathbf{s} \rangle_{\min}$	$\langle \mathbf{s} \rangle_{\max} / \langle \mathbf{s} \rangle_{\min}$
Thiers13	2.01	1.02	1.15
LyonSchool	2.03	1.04	1.24
SFHH	2.08	1.05	1.20
LH10	2.03	1.05	1.29
Mid1	2.08	1.05	1.27
F&F 2011-05	2.91	1.14	1.38
F&F 2010-08	2.83	1.21	1.57
F&F 2011-01	2.62	1.24	1.67
F&F 2010-10	3.20	1.29	1.73
F&F 2011-03	2.75	1.29	1.75
Kenyan	2.72	1.33	1.96
F&F 2010-12	3.36	1.36	1.81
F&F 2010-11	3.32	1.39	1.90
F&F 2010-09	3.04	1.39	1.80
F&F 2011-04	2.93	1.41	2.00
F&F 2011-02	3.19	1.44	2.00
Copenhagen	2.72	1.82	2.66
HS11	9.98	1.93	3.29
HS12	9.85	1.99	3.44
Malawi	4.50	2.00	3.78
InVS15	6.34	3.23	5.25
InVS13	3.75	4.09	8.07

TABLE S1: For static hypergraphs, the average path size $\langle s \rangle_{\min}$ when choosing the minimum size (column 1) and the relative increase when selecting instead the mean (column 2) or maximum (column 3) hyperedge size.

Next we move to the temporal setting. We plot in Figure S7 the average path size as a function of the fastest path durations for all datasets to complete Figure 5c of the main text. The figure presents average path size using 3 different hyperedge selection strategies, and we shade in one standard deviation for each. There is now much greater variability, although the different strategies exhibit very similar behaviour across all datasets. We also observe that for some datasets path size is decreasing as path duration increases, whereas for others we observe path size remaining more or less constant across different path duration lengths.

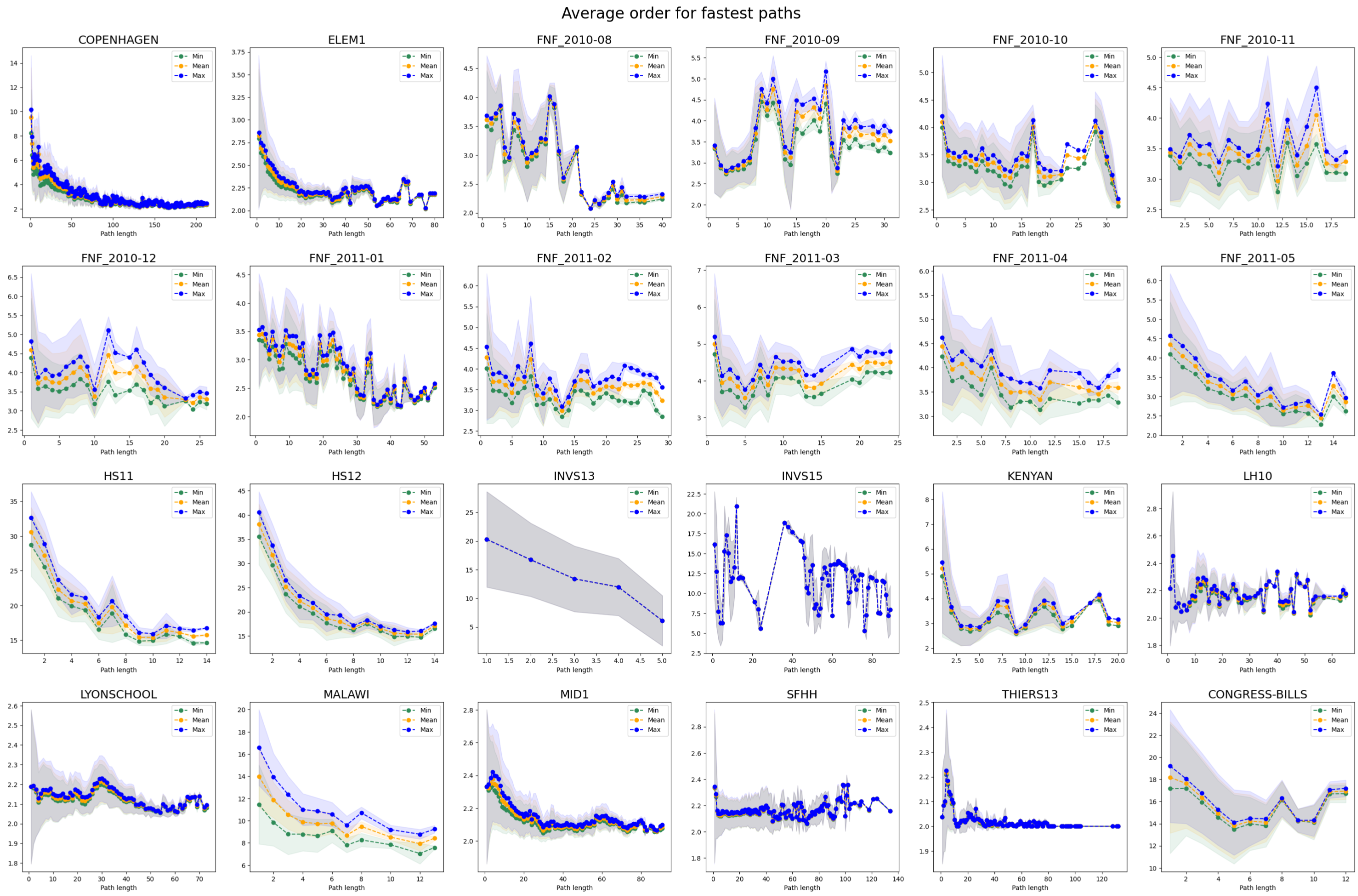


FIG. S7: Average interaction size $\langle s \rangle$ of the higher-order fastest paths as a function of the path duration for all datasets when using 3 different strategies (minimum in green ($\langle s \rangle_{\min}$), mean in orange $\langle s \rangle_{\text{mean}}$, maximum in blue $\langle s \rangle_{\max}$) to resolve ambiguities.

Finally, in Table S2 we provide the relative increase in average path size for temporal hypergraphs. As before, we first calculate $\langle s \rangle_{\min}$ in column 1 and then determine how much longer average path size would be if we employ alternative strategies by calculating the proportion $\langle s \rangle_{\max}/\langle s \rangle_{\min}$ (column 2) and then $\langle s \rangle_{\text{mean}}/\langle s \rangle_{\min}$ (column 3). In the temporal setting, the hyperedge selection strategy makes very little difference to the calculated average path size. This is because temporal hypergraphs don't have very many instances where multiple hyperedges are available at the same link, a feature we investigate next.

Datasets	$\langle \mathbf{s} \rangle_{\text{min}}$	$\langle \mathbf{s} \rangle_{\text{mean}} / \langle \mathbf{s} \rangle_{\text{min}}$	$\langle \mathbf{s} \rangle_{\text{max}} / \langle \mathbf{s} \rangle_{\text{min}}$
Thiers13	2.07	1.00	1.00
SFHH	2.14	1.00	1.00
LyonSchool	2.14	1.00	1.01
LH10	2.17	1.00	1.01
Elem1	2.35	1.02	1.04
F&F 2011-01	3.04	1.04	1.08
F&F 2010-09	3.14	1.04	1.07
F&F 2010-10	3.25	1.04	1.07
F&F 2010-11	3.29	1.05	1.09
F&F 2010-08	3.32	1.02	1.04
Kenyan	3.34	1.04	1.08
F&F 2011-05	3.36	1.06	1.12
F&F 2011-02	3.43	1.07	1.13
F&F 2010-12	3.62	1.06	1.12
F&F 2011-04	3.72	1.07	1.13
F&F 2011-03	3.76	1.08	1.14
Malawi	10.01	1.20	1.40
InVS15	11.21	1.00	1.00
Congress bills	16.31	1.03	1.06
InVS13	16.34	1.00	1.00
HS11	23.69	1.06	1.12
HS12	27.44	1.07	1.13

TABLE S2: For temporal hypergraphs, the average path size $\langle s \rangle_{\text{min}}$ when choosing the minimum size (column 1) and the relative increase when selecting instead the mean (column 2) or maximum (column 3) hyperedge size.

S4. HIGHER-ORDER PATH REDUNDANCY

In this section we study the amount of redundancy present in the systems when calculating average path size. To do this, we determine how often paths will have links with multiple hyperedges present. Table S3 compares the average number of such redundancies across all shortest paths in the static case with the average number of redundancies in the temporal setting. In each case, we calculate the average number of hyperedges present at each link in a path, along with the variability. While there is an extremely high amount of degeneracy in the static case, we observe that this disappears almost entirely in the temporal setting. For example, all static hypergraphs have on average more than 1 possible choice at each step along a path, and the Copenhagen dataset has on average 129.8 such choices. In contrast, more than half of the temporal hypergraphs have on average no redundancy at a step of the fastest path, and only one dataset, Thiers13, has more than 2. This means that in the static case we could expect to observe more variation in average path size for different hyperedge selection strategies, whereas for the temporal setting, all definitions would coincide much more closely.

Datasets	Static		Temporal	
	Mean	Std	Mean	Std
Copenhagen	129.81	271.91	1.27	1.62
F&F 2010-08	3.99	3.77	0.30	0.22
F&F 2010-09	10.80	12.24	0.90	0.26
F&F 2010-10	7.62	10.98	0.58	0.30
F&F 2010-11	13.36	18.75	0.41	0.34
F&F 2010-12	11.82	17.90	1.08	0.56
F&F 2011-01	6.023	8.90	0.41	0.34
F&F 2011-02	12.59	16.37	1.09	0.52
F&F 2011-03	11.78	17.12	1.11	0.54
F&F 2011-04	11.92	18.75	0.99	0.71
F&F 2011-05	4.93	5.29	0.76	0.68
HS11	30.02	33.54	0.84	0.60
HS12	22.34	32.17	0.53	0.49
InVS13	26.70	26.03	0.00	0.00
InVS15	32.98	29.66	0.00	0.00
Kenyan	9.32	9.84	0.36	0.32
LH10	3.63	4.13	0.08	0.05
LyonSchool	2.75	2.40	0.09	0.07
Malawi	84.82	112.87	2.24	1.76
Elem1	2.09	4.54	0.20	0.14
Mid1	2.83	3.15	0.001	0.003
SFHH	2.43	2.32	0.07	0.06
Thiers13	1.90	1.37	0.04	0.05
Congress-bills	44.48	38.52	0.18	0.37

TABLE S3: For static datasets, the average number of redundancies across all shortest paths, as well as the standard deviation.

To explore the temporal setting more fully, we extend Table S3 for the Copenhagen dataset by calculating 2 additional metrics. In Figure S8 the first panels presents the average number of redundancies for fastest paths. The second panel provides the proportion of steps where there is a redundancy present, given as a function of the path length. Similar analyses can be performed for all other datasets using the tools present in the hypergraphx [1] package. Together, these 2 figures indicate that longer paths tend to be those with less degeneracy, meaning that different definitions of average path size will converge, regardless the hyperedge selection strategy.

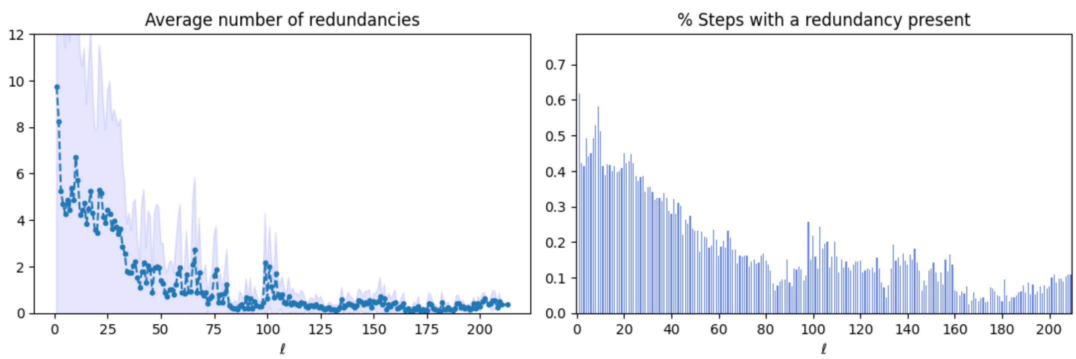


FIG. S8: For the Copenhagen dataset: Average number of redundancies for paths of each length (left) and the average percentage of steps in paths of specified length that have redundancies present. As can be seen, both are very low.

S5. DATA FROM A DIFFERENT DOMAIN: HYPERGRAPHS OF FROM SCIENTIFIC COLLABORATIONS

In this section, we analyse another system which is classically associated with higher-order interactions, but does not describe social interactions. To broaden our analysis, we present results for data originating from the domain of scientific collaborations [2, 3]. In Figure S9 we plot path length distributions for 4 static hypergraphs constructed from 4 different years of coauthorship data. Here one can clearly see the distribution shift even in the static case, because the static hypergraphs have not been constructed from the entire coauthorship dataset.

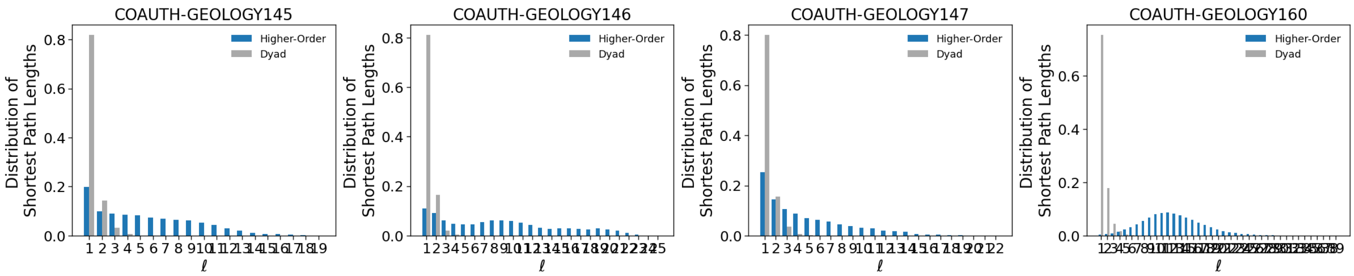


FIG. S9: Path length distributions for hypergraphs constructed from 4 different years of coauthorship data.

In Figure S10 we plot the average path size as a function of path length for the three strategies discussed in S3 and S4. As before, we observe that the behaviour of path size for different hyperedge-selection strategies is qualitatively very similar. Furthermore, average path size decreases for longer paths, although the 3rd frame exhibits an interesting increase for longer paths of around $\ell = 15$ steps.

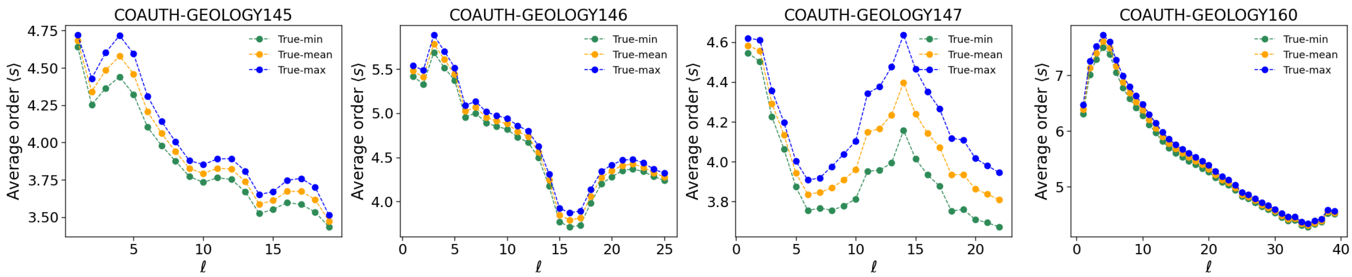
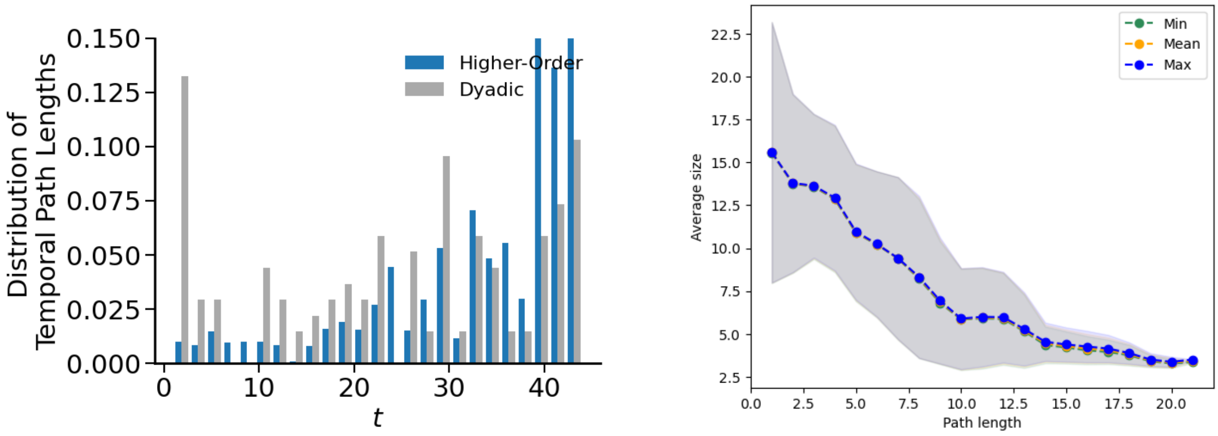


FIG. S10: Average path size $\langle s \rangle$ of higher-order shortest paths in coauthorship datasets as a function of the path length when using 3 different strategies (minimum in green ($\langle s \rangle_{\mathrm{min}}$), mean in orange $\langle s \rangle_{\mathrm{mean}}$, maximum in blue $\langle s \rangle_{\mathrm{max}}$) to resolve ambiguities.

We then move from the static to the temporal case and construct a temporal hypergraph network consisting of coauthorship data that spans 45 years and provide in Figure S11 the fastest path distributions and average fastest path size. For this dataset, the 3 different formulations of average path size in the temporal case coincide exactly, implying that there is no redundancy for this temporal dataset.

We investigate the presence of redundancy in the system in Figure S12 and see that indeed there is a very small average number of redundancies for shortest paths (always less than 0.2) and that a very small proportion of steps in hyperpaths have a redundancy present (less than 10% of paths have multiple hyperedges). Interestingly, here we observe a trend opposite to that of the previous figures: the average number of redundancies seems to be increasing with increasing path length, even though the increase is very slight. This makes sense for coauthorship data: it is not as common for authors to jointly co-author a small paper within the same year because it requires more work, but large papers



(a) (b)

FIG. S11: Fastest path length distribution (top) and average size (bottom) for a temporal network of coauthorship data spanning 46 years.

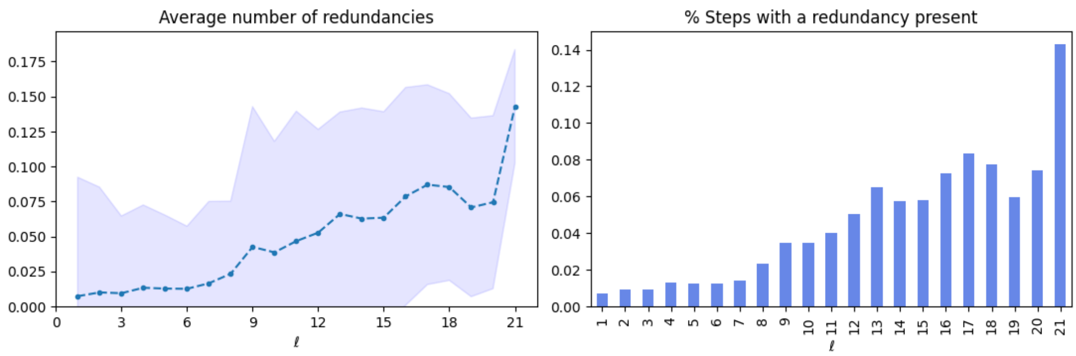


FIG. S12: For the coauthorship dataset: Average number of redundancies for paths of each length (left) and the average percentage of steps in paths of specified length that have redundancies present. As can be seen, both are very low.

[1] Quintino Francesco Lotito, Martina Contisciani, Caterina De Bacco, Leonardo Di Gaetano, Luca Gallo, Alberto Montresor, Federico Musciotto, Nicolò Ruggeri, and Federico Battiston. Hypergraphx: a library for higher-order network analysis. *Journal of Complex Networks*, 11(3):cnad019, 2023.

[2] Austin R. Benson, Rediet Abebe, Michael T. Schaub, Ali Jadbabaie, and Jon Kleinberg. Simplicial closure and higher-order link prediction. *Proceedings of the National Academy of Sciences*, 2018.

[3] Arnab Sinha, Zhihong Shen, Yang Song, Hao Ma, Darrin Eide, Bo-June (Paul) Hsu, and Kuansan Wang. An overview of microsoft academic service (MAS) and applications. In *Proceedings of the 24th International Conference on World Wide Web*. ACM Press, 2015.